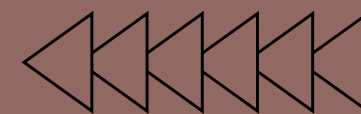


GRADE 11

GENERAL MATHEMATICS

Applications of Sequences and
Series in Financial Mathematics:



Term 1 Week 5

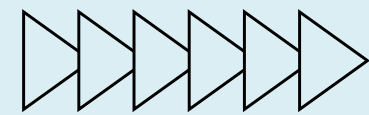
LEARNING COMPETENCIES



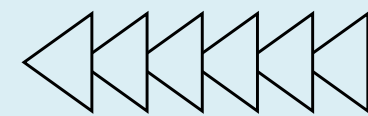
- Apply sequences and series to solve financial problems, such as periodic payments for loans and mortgages.



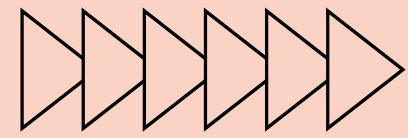
- Determine the accumulated values of investment over time using sequences and series.



Lesson 3. Applications of Sequences and Series in Financial Mathematics: Understanding Appreciation, Depreciation, Compound Interest, Loans, Mortgage Payment and Investment Growth



ACTIVITY A.1: MY ONE MILLION JOURNEY



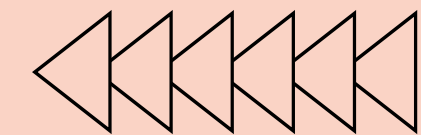
INSTRUCTIONS:

Read and analyze the given scenario and answer the questions that follow.

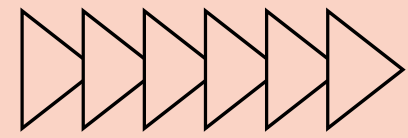
Scenario:

Imagine it is 10 years from now. You have successfully graduated from college, started working, and built a good career. A bank is offering you a loan of ₱1 million. You now have an important decision to make. How would you use the money?

You may choose one of the following options:



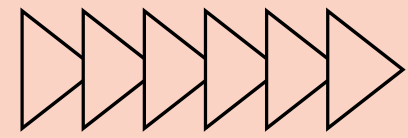
ACTIVITY A.1: MY ONE MILLION JOURNEY



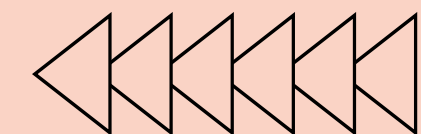
BUY A HOUSE



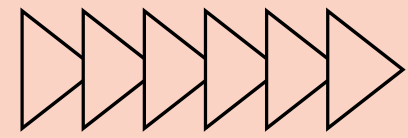
ACTIVITY A.1: MY ONE MILLION JOURNEY



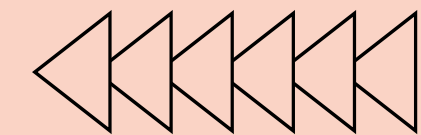
BUY A CAR



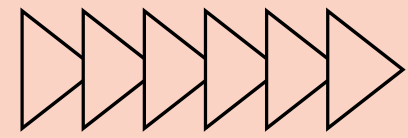
ACTIVITY A.1: MY ONE MILLION JOURNEY



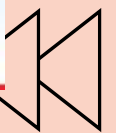
START A BUSINESS



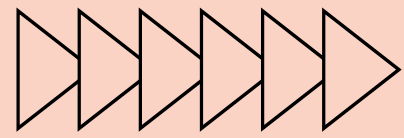
ACTIVITY A.1: MY ONE MILLION JOURNEY



INVEST IN LIFE INSURANCE



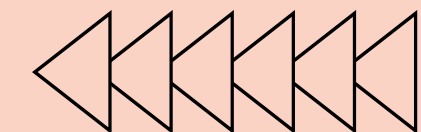
PROCESSING QUESTIONS:



INSTRUCTIONS:

Ten years from the date you acquired your property (chosen from above), you have fully paid off your loan. Now, you want to sell your item to invest in something bigger (e.g., new business, real estate, expansion).

- **What would you do next?**
- **How much do you think your asset is now worth?**
- **Was your original decision a smart financial choice? Why or why not?**
- **Do you think your property value would have increased or decreased after 10 years from the date you acquired your property? Why?**



**ACTIVITY A.2:
THE BEST
INVESTMENT**

INSTRUCTIONS:

Analyze the given set of problems and decide which is a better investment, one that follows an arithmetic sequence or one that follows a geometric sequence? Why?

PROBLEM #1

Arithmetic Growth

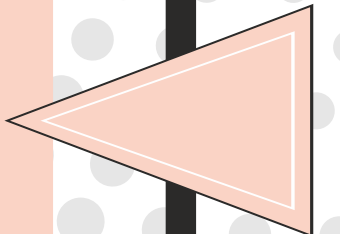
If the value of your chosen item increases by ₱50,000 every year, what would its value be after 10 years?

PROBLEM #2



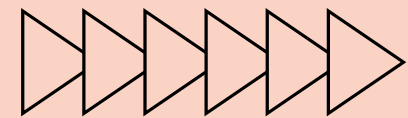
Geometric Growth

What if, instead of a fixed amount, your property's value increases by 5% each year? What would its value be after 10 years?



Activity B.1:
UNDERSTANDING
ESSENTIAL TERMS IN
FINANCIAL MATHEMATICS

ESSENTIAL TERMS IN FINANCIAL MATHEMATICS

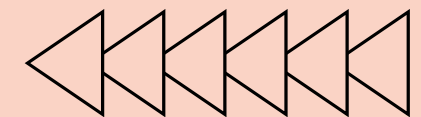


APPRECIATION

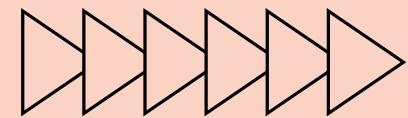
Appreciation is the increase in the value of an asset (like a house, land, or investment) over time. It often happens because of factors like demand, improvements, or inflation.

Example:

A lot owner bought a piece of land for ₱500,000. The value of the land appreciates by 8% every year..



ESSENTIAL TERMS IN FINANCIAL MATHEMATICS

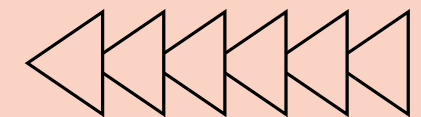


DEPRECIATION

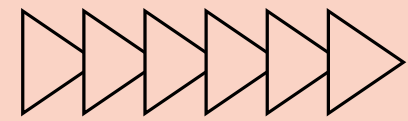
Depreciation is the decrease in the value of an asset over time. It usually happens due to usage, aging, or becoming outdated.

Example:

A car bought for ₱800,000 drops in value to ₱500,000 after several years..



ESSENTIAL TERMS IN FINANCIAL MATHEMATICS



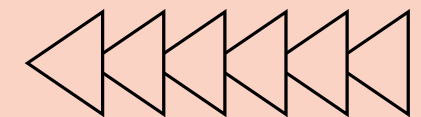
SIMPLE & COMPOUND INTEREST

Simple Interest - interest that is computed on the principal and then added into it.

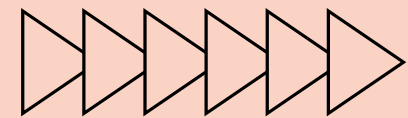
Compound Interest - interest is computed on the principal and on the accumulated past interest

Example:

Saving ₱10,000 with 5% interest compounded annually means you earn interest not just on ₱10,000, but also on the interest you already earned.



ESSENTIAL TERMS IN FINANCIAL MATHEMATICS

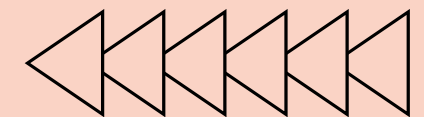


LOAN

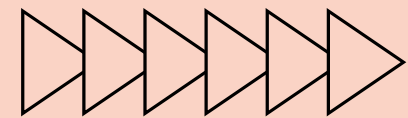
A **loan** is money borrowed from a bank or lender that you must pay back over time, usually with added interest.

Example:

Borrowing ₱1 million from the bank to buy a house and paying it back monthly with interest.



ESSENTIAL TERMS IN FINANCIAL MATHEMATICS

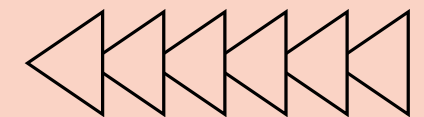


MORTGAGE

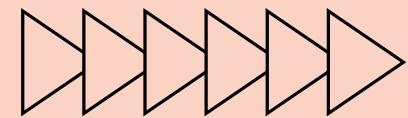
A **mortgage** is a specific type of loan used to buy real estate (like a house or land). The property itself is used as a guarantee (collateral) until the loan is fully paid.

Example:

You borrow money from a bank to buy a house, and if you don't pay, the bank can take the house.



ESSENTIAL TERMS IN FINANCIAL MATHEMATICS

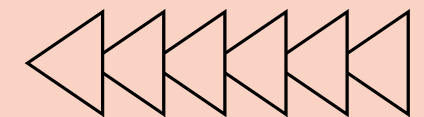


ACCUMULATED VALUE

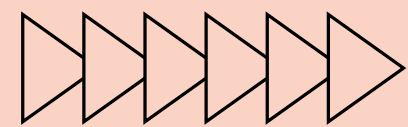
Accumulated value (or future value) of an investment is the total amount an investment is worth after a certain time, including the original investment and all growth (appreciation or interest).

Example:

If you invest ₱100,000 today and it earns ₱50,000 interest in 5 years, the accumulated value is ₱150,000.

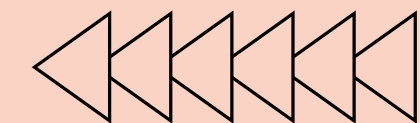


CONVERSION OR INTEREST PERIOD-TIME BETWEEN SUCCESSIVE CONVERSIONS OF INTEREST

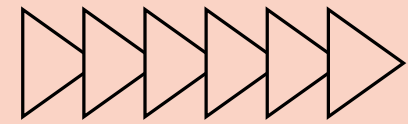


REMEMBER

- ❖ If compounded **annually** the conversion period is 1 year
- ❖ If compounded **quarterly** the conversion period is 3 months
- ❖ If compounded **monthly** the conversion period is 1 month
- ❖ If **semi-annually** the conversion period is 6 months
- ❖ **Frequency of conversion** -the number of conversion periods in
one year.

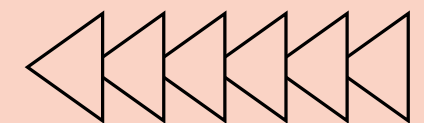


CONVERSION OR INTEREST PERIOD-TIME BETWEEN SUCCESSIVE CONVERSIONS OF INTEREST

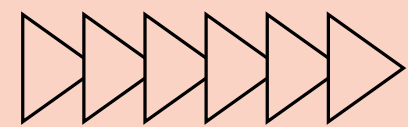


EXAMPLE

If an account earns 5% annual interest compounded quarterly, the frequency of conversion is 4 times a year.

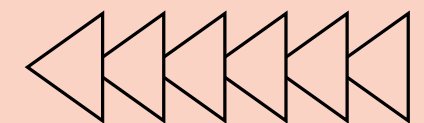


CONVERSION OR INTEREST PERIOD-TIME BETWEEN SUCCESSIVE CONVERSIONS OF INTEREST



REMEMBER

- ☐ **Rate** – yearly rate of increase of the investment
- ☐ **Period Rate** - the rate of interest for one conversion period
- ☐ **Time of term** - how long in years the money is borrowed or invested.



**ACTIVITY B.2:
APPLICATION OF
SEQUENCE AND
SERIES IN
FINANCIAL
PROBLEMS**

INSTRUCTIONS:

Financial problems, such as periodic payments for loans and mortgages can be solved using sequences and series. These mathematical tools are essential in determining the accumulated value of investments over time. Below are some examples that illustrate these applications. Participate in the discussion for better understanding.

PROBLEM #1

A lot owner bought a piece of land for ₱500,000. The value of the land appreciates by 8% every year. What will be the value of the land after 5 years?

PROBLEM #2



A cell phone was purchased for ₱32,500 in 2015. If the value of the cell phone depreciated annually by 16%. Find its value in 2025

PROBLEM #3

₱3000 is placed in an account at 5% interest compounded quarterly for 5 years. How much is in the account at the end of 5 years?

Compound interest calculations take place every conversion period. Therefore, when using sequences, we take the number of conversion periods as our counter.

PROBLEM #4

Christina decides to save money for after graduation. Christina starts by setting aside ₱100. Each week, Christina increases the amount she saves by ₱20. How much money will Christina have saved after 52 weeks?

PROBLEM #5

A savings account has an annual interest rate of 3% compounded monthly. Find the amount in the account 18 months after ₱5,000 is invested.

THANK YOU

